

VIANKA VANESSA CASTRO ORDOÑEZ

Guatemala City, Guatemala | +502 5512 5234 | vianka.v.castro@gmail.com | <https://github.com/Vann06>

EDUCATION

Universidad del Valle de Guatemala

Guatemala City, Guatemala

Jan. 2023 - Present

B.S. in Computer Science and Information Technology; 8th semester

Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Databases, Software Engineering, Artificial Intelligence, Operating Systems, Compiler/Automata Theory

Colegio Decroly Americano

Guatemala City, Guatemala

2011 - 2022

Bachelor of Science and Letters; High Honors recognition

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, PHP, Kotlin, C, SQL, HTML, CSS, Rust

Frameworks & Web: Laravel, Vue.js, React, Node.js, Vite, REST APIs, MVC

Databases & Cloud: PostgreSQL, SQL Server, SQLite, Firebase Firestore, AWS, Azure fundamentals

Tools & Practices: Git, GitHub, Docker, Docker Compose, Jira/Atlassian, CI/CD fundamentals, Agile/Scrum, Visual Studio, IntelliJ, Android Studio, LaTeX, MOS Excel

Languages: Spanish: Native; English: Upper-intermediate (B2)

PROFESSIONAL EXPERIENCE

Computer Science Teaching Assistant

Jun. 2025 - Present

Universidad del Valle de Guatemala

- Support students in Object-Oriented Programming and Algorithms/Dynamic Programming through practice sessions, debugging guidance, and code review.
- Explain concepts such as classes, inheritance, modular design, algorithmic thinking, recurrence, and dynamic programming in clear, practical terms.
- Review assignments and provide structured feedback on correctness, readability, and problem-solving approach.

Mathematics Teaching Assistant

Jan. 2023

Universidad del Valle de Guatemala

- Tutor university-level students in Discrete Mathematics and Probability Theory, including logic, set theory, relations, combinatorics, probability distributions, expected value, and structured problem solving.
- Prepare guided exercises and exam-review sessions that help students translate abstract mathematical concepts into clear, step-by-step solutions

RELEVANT SKILLS

- Native Spanish speaker and fluent in English
- Strong time management and leadership abilities
- Proficient with computers and quick to adapt to new software and programs
- Skilled at integrating into and contributing to new teams

SELECTED PROJECTS

Sol Store E-commerce Web Application

[GitHub](#)

Technologies: Laravel, Vue.js, PHP, PostgreSQL, Docker, Docker Compose, JavaScript

- Developed a full-stack online store for personalized 3D-printed products with frontend, backend, database, and containerized development environment.
- Implemented product management, user-facing flows, API configuration, and analytics support for tracking e-commerce behavior.

Medical Robot Decision-Making Simulation

[GitHub](#)

Technologies: Python, Jupyter Notebook, Value Iteration, Q-Learning, GridWorld

- Compared model-based and learning-based approaches for a stochastic hospital robot using shared states, actions, rewards, and performance metrics.

Lexer Generator

[GitHub](#)

Technologies: Rust, Automata Theory, Regular Expressions, DFA/NFA, Transition Tables

- Built a lexical analyzer generator from YALex specifications using regex parsing, Thompson NFA construction, DFA conversion, minimization, and simulation.

TUTO! Tutoring Management App

[GitHub](#)

Technologies: Kotlin, Android Studio, Firebase Authentication, Firebase Firestore, Jetpack Compose

- Built an Android app for individual and group tutoring management, including authentication, scheduling, user profiles, dark/light mode, and Firestore persistence.

LEADERSHIP & INVOLVEMENT

- Recognized as a Distinguished Student at Universidad del Valle de Guatemala for academic performance.
- Strengthened communication and leadership skills through academic tutoring and collaborative software projects.